

Optimal shrinkage in a simple hierarchical model

BIOS/BCB 784, companion to hier/hier_simulation.R

Setup

For genes $i = 1, \dots, m$ and replicates $j = 1, \dots, n$:

$$\theta_i \sim N(\mu_\theta, \sigma_\theta^2), \quad x_{ij} \mid \theta_i \sim N(\theta_i, \sigma_x^2). \quad (1)$$

Let $\bar{x}_i = \frac{1}{n} \sum_j x_{ij}$. We consider the shrinkage estimator

$$\hat{\theta}_i = (1 - B)\bar{x}_i + B\mu_\theta, \quad B \in [0, 1], \quad (2)$$

where μ_θ , σ_θ^2 and σ_x^2 are treated as known.

Question

Work out the value of B that minimizes the mean squared error $E[(\hat{\theta}_i - \theta_i)^2]$, averaging over both θ_i and the data.

Answer

Given θ_i , the average of n independent draws has

$$\bar{x}_i \mid \theta_i \sim N(\theta_i, \sigma^2), \quad \text{where} \quad \sigma^2 = \frac{\sigma_x^2}{n}. \quad (3)$$

Adding and subtracting θ_i ,

$$\hat{\theta}_i - \theta_i = (1 - B)(\bar{x}_i - \theta_i) + B(\mu_\theta - \theta_i). \quad (4)$$

Let's call the two pieces $e_i = \bar{x}_i - \theta_i$ (the measurement error) and $d_i = \mu_\theta - \theta_i$ (how far the gene's true value is from the population mean). Squaring the expression above gives

$$(\hat{\theta}_i - \theta_i)^2 = (1 - B)^2 e_i^2 + B^2 d_i^2 + 2B(1 - B)e_i d_i, \quad (5)$$

and we need the expected value of each of the three terms.

- The measurement error e_i is $N(0, \sigma^2)$ no matter what value θ_i takes. So $E[e_i] = 0$ and $E[e_i^2] = \text{Var}(e_i) = \sigma^2$.
- The gene's deviation d_i comes from the prior, $\theta_i \sim N(\mu_\theta, \sigma_\theta^2)$. So $E[d_i] = 0$ and $E[d_i^2] = \sigma_\theta^2$.
- The measurement error does not depend on which θ_i the gene has, so e_i and d_i are independent. Then $E[e_i d_i] = E[e_i]E[d_i] = 0$, and the cross term drops out.

Putting these together:

$$\text{MSE}(B) = E[(\hat{\theta}_i - \theta_i)^2] = (1 - B)^2 \sigma^2 + B^2 \sigma_\theta^2. \quad (6)$$

The first term is the variance of the estimator and the second term is its average squared bias. Increasing B lowers the variance but raises the bias, so the best choice is somewhere in between. This is the same

bias-variance trade-off that appears in ridge regression, smoothing, and many other settings where we accept a little bias in exchange for a large reduction in variance. Taking the derivative,

$$\frac{d \text{MSE}}{dB} = -2(1 - B)\sigma^2 + 2B\sigma_\theta^2 = 0, \quad (7)$$

and solving for B ,

$$B^* = \frac{\sigma^2}{\sigma^2 + \sigma_\theta^2}. \quad (8)$$

To interpret B^* , look at the variance of the observed averages. We can write $\bar{x}_i = \theta_i + e_i$, the true value plus measurement error. These two pieces are independent, so their variances add:

$$\text{Var}(\bar{x}_i) = \text{Var}(\theta_i) + \text{Var}(e_i) = \sigma_\theta^2 + \sigma^2. \quad (9)$$

The denominator of B^* is exactly this total variance, and the numerator is the part of it due to measurement error:

$$B^* = \frac{\sigma^2}{\text{Var}(\bar{x}_i)} = \frac{\text{noise variance}}{\text{total variance}}. \quad (10)$$

So B^* is the fraction of the spread in the $\bar{x}_i$ that is noise. If most of the differences between genes' averages are just measurement error, the $\bar{x}_i$ are unreliable and we should shrink them strongly toward μ_θ . If most of the differences reflect real differences in θ_i , we should mostly trust each gene's own average.

With the parameters in the script ($n = 5$, $\sigma_x^2 = 4$, $\sigma_\theta^2 = 1.5$), we have $\sigma^2 = \frac{4}{5} = 0.8$ and

$$B^* = \frac{0.8}{0.8 + 1.5} = 0.348, \quad (11)$$

which is close to the minimum found by grid search in the simulation. Plugging B^* back into Equation 6 gives the minimum MSE,

$$\text{MSE}(B^*) = \frac{\sigma_\theta^2 \sigma^2}{\sigma_\theta^2 + \sigma^2} = \frac{1.2}{2.3} = 0.522. \quad (12)$$

Compare this to the alternatives:

- No shrinkage ($B = 0$): $\text{MSE}(0) = \sigma^2 = 0.8$.
- Complete shrinkage ($B = 1$): $\text{MSE}(1) = \sigma_\theta^2 = 1.5$.
- The variance of the data, $\text{Var}(\bar{x}_i) = \sigma_\theta^2 + \sigma^2 = 2.3$.

The optimal MSE is smaller than both σ^2 and σ_θ^2 : it is one half of their harmonic mean, so its inverse is the sum of the two precisions, $1/\text{MSE}(B^*) = 1/\sigma^2 + 1/\sigma_\theta^2$. Equivalently, $\text{MSE}(B^*) = \sigma_\theta^2 \sigma^2 / \text{Var}(\bar{x}_i)$, the product of the signal and noise variances divided by the total variance of the $\bar{x}_i$.